

```

> use college
< switched to db college
> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      averageQty: { $avg: '$quantity' },
    },
  },
]);
< [
  {
    _id: 'Lattes',
    averageQty: 27.5
  },
  {
    _id: 'Americanos',
    averageQty: 17.5
  },
  {
    _id: 'Cappuccino',
    averageQty: 16.333333333333332
  },
  {
    _id: 'Mochas',
    averageQty: 11
  }
]

```

```

> db.sales.aggregate([
  {
    $match: {
      item: "Americanos"
    }
  },
  {
    $group: {
      _id: "$item",
      averageQty: { $avg: "$quantity" }
    }
  }
]);
< [
  {
    _id: 'Americanos',
    averageQty: 17.5
  }
]
> db.sales.aggregate([
  {
    $match: {
      size: "Grande"
    }
  },
  {
    $group: {
      _id: "$size",
      averageQty: { $avg: "$quantity" }
    }
  }
]);
< [
  {
    _id: 'Grande',
    averageQty: 18.75
  }
]

```

```

> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      averageAmount: { $avg: { $multiply: ['$quantity', '$price'] } },
    },
  },
  { $sort: { averageAmount: 1 } },
])
< {
  _id: 'Cappuccino',
  averageAmount: 127.33333333333333
}
{
  _id: 'Americanos',
  averageAmount: 140
}
{
  _id: 'Mochas',
  averageAmount: 275
}
{
  _id: 'Lattes',
  averageAmount: 562.5
}
> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      averageAmount: { $avg: { $multiply: ['$quantity', '$price'] } },
    },
  },
  { $match: { averageAmount: { $gt: 150 } } },
  { $sort: { averageAmount: 1 } },
]);

```

```

> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      itemCount: { $count: {} },
    },
  },
])
< {
  _id: 'Mochas',
  itemCount: 1
}
{
  _id: 'Cappuccino',
  itemCount: 3
}
{
  _id: 'Americanos',
  itemCount: 4
}
{
  _id: 'Lattes',
  itemCount: 2
}

```

```

> db.sales.aggregate([
  {
    $group: {
      _id: null,
      totalQty: { $sum: '$quantity' },
    },
  },
]);
< [
  {
    _id: null,
    totalQty: 185
  }
]

```

```

> db.sales.aggregate([
  {
    $group: {
      _id: null,
      totalQty: { $sum: '$quantity' },
    },
  },
]

```

✖ • **SyntaxError:** Unexpected token (8:0)

```

    6 |   },
    7 |   },
>  8 |
      | ^
> db.sales.aggregate([
  {
    $group: {
      _id: null,
      totalQty: { $sum: '$quantity' },
    },
  },
  { $project: { _id: 0 } },
]);
< [
  {
    totalQty: 185
  }
]

```

```

> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      totalQty: { $sum: '$quantity' },
    },
  },
]);
< [
  {
    _id: 'Lattes',
    totalQty: 55
  },
  {
    _id: 'Americanos',
    totalQty: 70
  },
  {
    _id: 'Cappuccino',
    totalQty: 49
  },
  {
    _id: 'Mochas',
    totalQty: 11
  }
]

```

```
> db.sales.aggregate([
  {
    $group: {
      _id: null,
      maxQty: { $min: '$quantity' },
    },
  },
  {
    $project: {
      _id: 0,
    },
  },
]);
```

```
< {
  maxQty: 11
}
```

```
> db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      minQty: { $min: '$quantity' },
    },
  },
]);
```

```
< {
  _id: 'Cappuccino',
  minQty: 12
}
{
  _id: 'Americanos',
  minQty: 12
}
{
  _id: 'Lattes',
  minQty: 25
}
```

